Alex M. Haberlie, Ph.D.
Assistant Professor
Department of Earth, Atmosphere, and Environment
Northern Illinois University
DeKalb, IL, 60115
(815) 753-0631 – ahaberlie1@niu.edu

Education

2018 **Ph.D., Geography**
Northern Illinois University, DeKalb, IL
Dissertation: *Observed and Future Dynamically Downscaled Estimates of Precipitation Associated with Mesoscale Convective Systems*
Advisor: Dr. Walker Ashley

2014 **M.S., Geography**
Northern Illinois University, DeKalb, IL
Thesis: *Convective Initiation Climatology for the Atlanta, Georgia Region*
Advisors: Dr. Walker Ashley, Dr. Thomas Pingel

2010 **B.S., Computer Science**
University of Wisconsin – Platteville, Platteville, WI

Professional Appointments

2021 – Present **Assistant Professor**
Department of Earth, Atmosphere and Environment
Northern Illinois University, DeKalb, IL

2018 – 2021 **Assistant Professor**
Department of Geography and Anthropology
Louisiana State University, Baton Rouge, LA

2014 – 2018 **Instructor and Graduate Research Assistant**
Department of Geographic and Atmospheric Sciences
Northern Illinois University, DeKalb, IL

2012 – 2014 **Graduate Teaching Assistant**
Department of Geographic and Atmospheric Sciences
Northern Illinois University, DeKalb, IL

Research Interests

Observed and future projections of climate extremes, severe and hazardous weather, land use/land cover effects on regional climatology, applications of machine learning and computer vision.

Peer-reviewed Publications (**21 Publications; 10 first author**; **11 co-author**)

[21] Bhowmick, R., J. C. Trepanier, and **A. M. Haberlie**, 2023: Classification Analysis of Southwest Pacific Tropical Cyclone Intensity Changes Prior to Landfall. *Atmosphere*, **14**, 253. https://doi.org/10.3390/atmos14020253.

[20] Ashley, W S., **A. M. Haberlie**, and V. A. Gensini, 2023: The Future of Supercells in the United States. *Bulletin of the American Meteorological Society*, **104**, E1-E21. https://doi.org/10.1175/bams-d-22-0027.1.

[19] Wallace, B. C., **A. M. Haberlie**, W. S. Ashley, V. A. Gensini, and A. C. Michaelis, 2023: Decomposing the precipitation response to climate change in convection allowing simulations over the conterminous United States. Earth and Space Science, 10, e2023EA003094. https://doi.org/10.1029/2023ea003094.

[18] **Haberlie, A. M.**, W. S. Ashley, V. A. Gensini, and A. C. Michaelis, 2023: The ratio of mesoscale convective system precipitation to total precipitation increases in future climate change scenarios. *npj Climate and Atmospheric Science*, **6**, 150. https://doi.org/10.1038/s41612-023-00481-5.

[17] Gensini, V. A., **A. M. Haberlie**, and W. S. Ashley, 2023: Convection-Permitting Simulations of Historical and Possible Future Climate over the Contiguous United States. *Climate Dynamics*, **60**, 109-126. https://doi.org/10.1007/s00382-022-06306-0.

[16] Strader, S. M., W. S. Ashley, **A. M. Haberlie**, and K. Kaminski, 2022: Revisiting U.S. Nocturnal Tornado Vulnerability and its Influence on Tornado Impacts. *Weather, Climate, and Society.* **14**, 1147-1163. https://doi.org/10.1175/wcas-d-22-0020.1.

[15] Bhowmick, R., J. C. Trepanier, and **A. M. Haberlie**, 2022: Southwest Pacific Tropical Cyclone Development Classification Utilizing Machine Learning and Synoptic Composites. *International Journal of Climatology*, **42**, 4187-4213. https://doi.org/10.1002/joc.7457.

[14] **Haberlie, A. M.**, W. S. Ashley, C. M. Battisto, and V. A. Gensini, 2022: Thunderstorm Activity Under Intermediate and Extreme Climate Change Scenarios. *Geophysical Research Letters*, **49**, e2022GL098779. https://doi.org/10.1029/2022gl098779.

[13] Strader, S. M., **A. M. Haberlie**, and A. Loitz, 2021: Assessment of NWS County Warning Area Tornado Risk, Exposure, and Vulnerability. *Weather, Climate, and Society,* **12**, 189-209. https://doi.org/10.1175/wcas-d-20-0107.1.

[12] **Haberlie, A. M.**, W. S. Ashley, and M. Karpinski, 2021: Mean storms: Composites of radar reflectivity images during two decades of severe thunderstorm events. *International Journal of Climatology*, **41**, E1738-E1756. https://doi.org/10.1002/joc.6804.

[11] Ashley, W. S., **A. M. Haberlie**, and V. A. Gensini, 2020: Reduced frequency and size of late-twenty-first- century snowstorms over North America. *Nature Climate Change*, **10**, 539-544. https://doi.org/10.1038/s41558-020-0774-4.

[10] Gensini, V. A., **A. M. Haberlie**, and P. T. Marsh, 2020: Practically Perfect Hindcasts of Severe Convective Storms. *Bulletin of the American Meteorological Society*, **101**, E1259-E1278. https://doi.org/10.1175/bams-d-19-0321.1.

[9] Ashley, W. S., **A. M. Haberlie**, and J. Strohm, 2019: A Climatology of Quasi-Linear Convective Systems and Their Hazards in the United States. *Weather and Forecasting*, **34**, 1605-1631. https://doi.org/10.1175/waf-d-19-0014.1.

[8] **Haberlie, A. M.**, and W. S. Ashley, 2019: A radar-based climatology of mesoscale convective systems in the United States. *Journal of Climate*, **32**, 1591-1606. https://doi.org/10.1175/jcli-d-18-0559.1.

[7] **Haberlie, A. M.**, and W. S. Ashley, 2019: Climatological representation of mesoscale convective systems in a dynamically downscaled climate simulation. *International Journal of Climatology*, **39**, 1144- 1153. https://doi.org/10.1002/joc.5880.

[6] **Haberlie, A. M.**, and W. S. Ashley, 2018: A Method for Identifying Mesoscale Convective Systems in Radar Mosaics. Part I: Segmentation and Classification. *Journal of Applied Meteorology and Climatology*, **57**, 1575-1598. https://doi.org/10.1175/jamc-d-17-0293.1.

[5] **Haberlie, A. M.**, and W. S. Ashley, 2018: A Method for Identifying Midlatitude Mesoscale Convective Systems in Radar Mosaics. Part II: Tracking. *Journal of Applied Meteorology and Climatology*, **57**, 1599–1621. https://doi.org/10.1175/jamc-d-17-0294.1.

[4] **Haberlie, A. M.**, W. S. Ashley, A. Fultz, and S. Eagan, 2016: The effect of reservoirs on the climatology of warm-season thunderstorms in Southeast Texas, USA. *International Journal of Climatology*, **36**, 1808-1820. https://doi.org/10.1002/joc.4461.

[3] Ashley, W. S., S. Strader, D. Dziubla, and **A. M. Haberlie**, 2015: Driving blind: Weather-related vision hazards and fatal motor vehicle crashes. *Bulletin of the American Meteorological Society,* **96**, 755-778. https://doi.org/10.1175/bams-d-14-00026.1.

[2] **Haberlie, A. M.**, W. S. Ashley, and T. Pingel, 2015: The effect of urbanization on the climatology of thunderstorm initiation. *Quarterly Journal of the Royal Meteorological Society*, **141**, 663-675. https://doi.org/10.1002/qj.2499.

[1] **Haberlie, A. M.**, K. Gale, D. Changnon, and M. Tannura, 2014: Climatology of tropical system rainfall on the eastern Corn Belt. *Journal of Applied Meteorology and Climatology*, **53**, 395-405. https://doi.org/10.1175/jamc-d-13-0151.1.

Publications submitted for review

Haberlie, A. M., Wallace, B., W. S. Ashley, A. C. Michaelis, and V. A. Gensini, submitted: Mesoscale Convective System Activity in the United States Under Intermediate and Extreme Climate Change Scenarios. 1979–2021. *Climatic Change*.

Andrews, M., V. A. Gensini, **A. M. Haberlie**, and W. S. Ashley, A. C. Michaelis, M. Taszarek, submitted: Climatology of the Elevated Mixed Layer Over the Contiguous United States and Northern Mexico: 1979–2021. *Journal of Climate.*

Bundy, L., V. A. Gensini, W. S. Ashley, **A. M. Haberlie**, D. Changnon, submitted: Analysis of United States Crop Conditions: 1986 - 2022. *Agronomy Journal*.

Kaminski K., W. S. Ashley, **A. M. Haberlie**, and V. A. Gensini, submitted: Future Derecho Potential in the United States. *Journal of Climate*.

Stinnett, S., V. A. Gensini, **A. M. Haberlie,** A. C. Michaelis, and W. S. Ashley, submitted: Future Changes in the Climate of Extreme Daily Precipitation over the Contiguous United States from Convection-Permitting Simulations. *Journal of Applied Meteorology and Climatology*.

Zeeb, A. W., W. S. Ashley, **A. M. Haberlie**, V. A. Gensini, and A. C. Michaelis, submitted: Supercell Precipitation Contribution to the United States Hydroclimate. *International Journal of Climatology*.

Invited Presentations

Haberlie, A. M., 2022: Thunderstorm Activity Under Intermediate and Extreme Climate Change Scenarios. *Department of Earth, Atmosphere, and Environment Colloquium Series*, DeKalb, IL.

Haberlie, A. M., 2022: Thunderstorms in a changing climate. *NIUSTEAM: STEM Café*, DeKalb, IL.

Haberlie, A. M., 2021: Knowledge Discovery in Datasets of Observed and Simulated Thunderstorms. *Department of Geographic and Atmospheric Sciences Colloquium Series*, DeKalb, IL.

Haberlie, A. M., 2020: An analysis of half a million radar reflectivity snapshots centered on severe weather reports from 1996 to 2017. *Central Louisiana Chapter of the American Meteorological Society Meeting*, Baton Rouge, LA.

Haberlie, A. M., 2018: Classifying Rainfall Areas with Machine Learning. *2018 Unidata Users Workshop,* Boulder, CO.

Haberlie, A. M., 2018: The Past, Present, and Future of Organized Thunderstorms. *Department of Geography and Anthropology Friday Forum*, Baton Rouge, LA.

Haberlie, A. M., and D. Changnon, 2013: Climatology of tropical system rainfall on the eastern Corn Belt. *Chicago Chapter of the American Meteorological Society*, DeKalb, IL.

Grants

2021 – 2024 **National Oceanic and Atmospheric Administration (NOAA)**
The Vortex Southeast Project – $166,851
Proposal Title: *Faster, Clearer, Stronger Communication and Action: Building IWT and Vulnerable Resident Connections to Improve Severe Weather Literacy and Outcome*

2021 – 2023	**National Oceanic and Atmospheric Administration (NOAA)** **The Adaptation Sciences Program – $25,729** Proposal Title: *Planning a Flood Resilient Future for New Orleans, LA*
2019 – 2022	**National Science Foundation (NSF)** **Physical and Dynamic Meteorology; Climate and Large-Scale Dynamics; AGS-ATM & Geospace Sciences – $60,382** Proposal Title: *Observed and Future Dynamically Downscaled Estimates of Precipitation Associated with Mesoscale Convective Systems*
2019 – 2021	**Louisiana Board of Regents (BoR)** **Research Competitiveness Subprogram – $86,157** Proposal Title: *Building a Climatology of Mesoscale Convective Processes Using Image-Classification and Machine- Learning Techniques on Sequences of Radar Data*
2018 – 2020	**National Oceanic and Atmospheric Administration (NOAA)** **National Sea Grant College Program – $200,000** Proposal Title: *Communicating Climate Tools to Coastal Stakeholders*
2014	**United States Geological Survey (USGS)** **Illinois Water Resources Center – $8,663** Proposal Title: *Hydro-meteorological responses to tropical system precipitation in Illinois*

Conference presentations (*graduate student advisee)

Corner, J. M.*, **A. M. Haberlie**, and S. C. Collis, 2023: Verification of Storm Mode in Model Using Machine Learning. *22nd Annual Student Conference*, Denver, CO, S13, https://ams.confex.com/ams/103ANNUAL/meetingapp.cgi/Paper/422811.

Haberlie, A. M., W. S. Ashley, V. A. Gensini, and A. Michaelis, 2023: Mesoscale Convective System Activity Under Intermediate and Extreme Climate Change Scenarios. *36th Conference on Climate Variability and Change*, Denver, CO, American Meteorological Society, 4C.1, https://ams.confex.com/ams/103ANNUAL/meetingapp.cgi/Paper/419605.

Haberlie, A. M., W. S. Ashley, V. A. Gensini, and A. Michaelis, 2022: Mesoscale Convective System Activity Under Intermediate and Extreme Climate Change Scenarios. *30th Conference on Severe Local Storms*, Santa Fe, NM, American Meteorological Society, 188, https://ams.confex.com/ams/30SLS/meetingapp.cgi/Paper/407719.

Haberlie, A. M., W. S. Ashley, V. A. Gensini, and A. Michaelis, 2022: Projected Changes in Mesoscale Convective Systems over North America. *19th Conference on Mesoscale Processes*, Virtual Meeting, American Meteorological Society, 14.3A, https://ams.confex.com/ams/102ANNUAL/meetingapp.cgi/Paper/395946.

Haberlie, A. M., W. S. Ashley, V. A. Gensini, and C. Battisto, 2021: Performance of Continental-Scale Regional Climate Simulations for High-Impact Weather Events. *34th Conference on Climate Variability and Change*, Virtual Meeting, American Meteorological Society, 14B.10, https://ams.confex.com/ams/101ANNUAL/meetingapp.cgi/Paper/384395.

Haberlie, A. M., W. S. Ashley, V. A. Gensini, and M. Karpinski, 2021: SVRIMG: Radar Reflectivity Images Centered on Severe Weather Reports. *11th Symposium on Advances in Modeling and Analysis Using Python*. American Meteorological Society, Virtual Meeting, 1037, https://ams.confex.com/ams/101ANNUAL/meetingapp.cgi/Paper/384398.

Haberlie, A. M., W. S. Ashley, V. A. Gensini, and M. Karpinski, 2020: Analysis and Application of Mesoscale Radar Scenes during Severe Weather Events. *19th Conference on Artificial Intelligence for Environmental Science*. Boston, MA, American Meteorological Society, 4.4, https://ams.confex.com/ams/2020Annual/meetingapp.cgi/Paper/368779.

Haberlie, A. M., and W. S. Ashley, 2019: Using Machine Learning Techniques to Construct a Climatology of Mesoscale Convective Systems in the United States. *18th Conference on Artificial and Computational Intelligence and Its Application to the Environmental Sciences*, Phoenix, AZ, American Meteorological Society, 183, https://ams.confex.com/ams/2019Annual/meetingapp.cgi/Paper/352493.

Haberlie, A. M., and W.S. Ashley, 2018: Climatological Representation of Mesoscale Convective Systems in a Dynamically Downscaled Climate Simulation, 2nd Convection-Permitting Climate Modeling Workshop, Boulder, CO, GEWEX, https://drive.google.com/file/d/1AlgM-WQ8GpMw2wQJlgg9gICJZNKG8-IZ/view

Haberlie, A. M., W. S. Ashley, and J. Strohm, 2018: A Climatology of Quasi-linear Convective Systems in the U.S. *2018 SWAAG Meeting*. Baton Rouge, LA, American Association of Geographers, http://www.sw-aag.org/uploads/2/4/4/5/24451940/2018_swaag_annual_meeting_-_abstracts.pdf.

Haberlie, A. M., and W. S. Ashley, 2018: Using Scikit-Learn to Increase the Precision of an Automated Mesoscale Convective System Segmentation and Tracking Procedure. *8th Symposium on Advances in Modeling and Analysis Using Python*. Austin, TX, American Meteorological Society, 281, https://ams.confex.com/ams/98Annual/webprogram/Paper327997.html.

Haberlie, A. M., and R. May, 2017: Implementation and Comparison of Interpolation Techniques for the Meteorological Python (MetPy) Package, *7th Symposium on Advances in Modeling and Analysis Using Python*. Seattle, WA, American Meteorological Society, 317, https://ams.confex.com/ams/97Annual/webprogram/Paper304950.html.

Haberlie, A. M., and W. S. Ashley, 2017: A Comparison of Machine Learning Approaches for Classification of Radar-derived Convective Clusters, 7th Symposium on Advances in Modeling and Analysis Using Python, Seattle, WA, American Meteorological Society, 318, https://ams.confex.com/ams/97Annual/webprogram/Paper308024.html.

Haberlie, A. M., and W. S. Ashley, 2016: A U.S. climatology of mesoscale convective systems: 1997- 2013. 15th Annual Student Conference, New Orleans, LA, American Meteorological Society, S81, https://ams.confex.com/ams/96Annual/webprogram/Paper292206.html.

Haberlie, A. M., and W. S. Ashley, 2014: A U.S. climatology of mesoscale convective systems: 1997- 2013. 27th Conference on Severe Local Storms, Madison, WI, American Meteorological Society, 143, https://ams.confex.com/ams/27SLS/webprogram/Paper256044.html.

Haberlie, A. M., W. S. Ashley, and T. J. Pingel, 2014: Comparison of Methodologies for Detecting Convective Initiation Due to Differences in Land Use. *11th Symposium on the Urban Environment*, Atlanta, Georgia, American Meteorological Society, 287, https://ams.confex.com/ams/94Annual/webprogram/Paper235719.html.

Honors and Awards

2017 – 2018	**Graduate Research Fellowship** Dissertation Completion Fellowship
2015 – 2016	**Graduate Research Assistantship** Great Journeys Program
2015 – 2016	**Outstanding Thesis Award** Thesis Title: *Convective Initiation Climatology for the Atlanta, Georgia Region*
2013 – 2014	**Outstanding Graduate Student Award**

Professional Memberships

2012 – Present	**American Meteorological Society (AMS)**
2014 – Present	**American Association of Geographers (AAG)**
2014 – Present	**Gamma Theta Upsilon (GTU)**

Professional Development

2023	**Unidata Users Workshop** University Corporation for Atmospheric Research Boulder, CO
2018	**Working Group Meeting** South Central Climate Adaptation Science Center Fort Worth, TX
2018	**2nd Convection-Permitting Climate Modeling Workshop** Global Energy and Water Exchanges Program (GEWEX) Boulder, CO
2018	**Unidata Users Workshop** University Corporation for Atmospheric Research Boulder, CO
2018	**Geography Faculty Development Alliance (GFDA)** **Early Career Faculty Workshop** Association of American Geographers Washington, D.C.
2016	**Unidata Intern** University Corporation for Atmospheric Research, Boulder, CO

Professional Activities

2021 – Present	**Associate Editor** *Artificial Intelligence for the Earth Systems* American Meteorological Society
2020 – Present	**Representative** Unidata Users Committee University Corporation for Atmospheric Research
2021	**Associate Editor** *Monthly Weather Review* American Meteorological Society
2018 – 2021	**Affiliate Member** Louisiana State University South Central Climate Adaptation Science Center

Professional Service

2023	**Co-Chair of Organization Committee** 2023 Unidata Users Workshop
2022	**Review Panelist** Unidata Equipment Awards
2020	**Session Co-Chair** 19th Conference on A.I. for Environmental Sci. American Meteorological Society
2019	**External reviewer** Geography and Spatial Sciences Program National Science Foundation
2018	**Local Host Committee Member** Southwest Division American Association of Geographers
2013 – 2023	**Peer Reviewer** *Bulletin of the American Meteorological Society, Nature Climate Change, Artificial Intelligence for the Earth Systems, Journal of Geophysical Research – Atmospheres, Journal of Atmospheric and Oceanic Technology, Geophysical Research Letters, International Journal of Climatology, Journal of Climate, Remote Sensing, Remote Sensing Letters, Journal of Applied Meteorology and Climatology, Journal of Hydrometeorology, Monthly Weather Review, Weather and Forecasting, and Urban Climate.*

Departmental Service

Northern Illinois University
Department of Earth, Atmosphere, and Environment
Previously: Department of Geographic and Atmospheric Sciences (2021 – 2022)

2023 – Present	**Undergraduate Advisor** Climate Science Emphasis
2022 – Present	**Faculty Advisor** American Meteorological Society Student Chapter
2022 – Present	**Committee Member** Infrastructure Committee

2021 – Present	**Booth Volunteer** STEMfest
2022	**Chair** Postdoctoral Researcher Search Committee

Louisiana State University
Department of Geography and Anthropology

2020 – 2021	**Undergraduate Coordinator** Geography B.A. / B.S.
2020 – 2021	**Committee Member** Speakers Committee
2018 – 2020	**Committee Member** Technology Committee

Mentorship

Postdoctoral Advising

2022 – Present	Brendan Wallace

M.S. / Ph.D. Advising

M.S. / Ph.D. Committee Member

2023	Daniel Curtis (NIU, M.S.)
2023	Aaron Zeeb (NIU, M.S.)
2023	Margo Andrews (NIU, M.S.)
2023	Sylvia Stinnett (NIU, M.S.)
2023	Kris Kaminski (NIU, M.S.)
2022	Michael Broussard (LSU, M.S)
2021	Chris Battisto (NIU, M.S.)
2021	Ashley Autore (LSU, M.S.)
2020	Marisa Karpinksi (LSU, M.S.)
2020	Bienvenu Massamba (LSU; M.S.)
2020	Rupsa Bhowmick (LSU, Ph.D.)

Undergraduate Independent Study

2023	Rachel Schmitt; Analyzing tornadoes and storm mode classifications in a Jupyter Notebook

Instructor of Record History (Combined graduate and undergraduate sections in bold)

	2014	2015	2016	2017
Spring		a	*	a, b
Fall	a	*	a, b	*

	2018	2019	2020
Spring	*	c, **e**	c, **g**
Fall	c, **d**	c, f	c, **h**

	2021	2022	2023
Spring	c, **e**	**j**, **	**j**, **
Fall	i,**	**k**, **	**k**, **

a: Northern Illinois University, Introduction to the Atmosphere (GEOG 105)
b: Northern Illinois University, Practicum in Weather Analysis and Forecasting (MET 475)
c: Louisiana State University, Physical Geography: The Atmosphere (GEOG 2050)
d: Louisiana State University, Special Topics in Geography (GEOG 4997)
e: Louisiana State University, Methods of Climatological Analysis (GEOG 4016)
f: Louisiana State University, Climatology (GEOG 4014)
g: Louisiana State University, Physical Climatology (GEOG 4015)
h: Louisiana State University, Hazard Risk Reduction (GEOG 4200)
i: Northern Illinois University, Severe and Hazardous Weather (GEOG 306)
j: Northern Illinois University, Computer Programming for the Geosciences (GEOG 493/593)
k: Northern Illinois University, Regional/Advanced Climatology (GEOG/EAE 370/670/671)

*: Graduate Research Assistantship
**: Externally Funded Course Release

Updated on January 3rd, 2024